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What is JavaScript?

In this article
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Summary

JavaScript (JS) is a lightweight interpreted programming language that allows you to implement complex features on web pages — every time a web page does more than just sit there and display static information for you to look at, JavaScript is probably involved. It often plays a part in a web page alongside the other technologies we've been learning about in this course — CSS and HTML. We've also been learning about client-side web APIs, and third-party APIs that are based on JavaScript. We've also been learning about how to use JavaScript to store and retrieve data, and how to create a basic server-side JavaScript application.

Learning resources

This course provides a set of learning modules, which build a complete beginner's course on JavaScript, starting with the most basic concepts and moving on to more complex topics.

A high-level definition

JavaScript is a cross-platform, object-oriented scripting language that allows you to implement complex features on web pages — every time a web page does more than just sit there and display static information for you to look at, JavaScript is probably involved. It is a lightweight, interpreted programming language, with object-oriented capabilities. It's dynamic, weakly typed, prototype-based, and has first-class functions. It's used for many non-browser environments, too, such as Node.js, Apache CouchDB, and Adobe Acrobat. It is a multi-paradigm language, supporting object-oriented, imperative, and functional programming styles. It's also used in game development, and in the creation of browser add-ons and user scripts. JavaScript is used in many of the most popular web-based projects, such as Google Maps, Facebook, and Twitter. It is also used in many non-web projects, such as Node.js, Electron, and Cordova.

- HTML is the markup language that we use to structure and give meaning to our web content, for example defining paragraphs, headings, and tables, or embedding images and interactive media-rich elements.
- CSS is the language we use to apply style to our markup, for example setting fonts and colors, adding borders and shadows, and defining how our content should be laid out on the page — for example, we can make it into a single column, or three columns, with a gutter between the first and second columns, if we want to. CSS can also be used to add some simple effects to our content, such as a drop shadow, or a transition when a link is hovered. CSS is a styling language, and it is widely used when you are working with a rich feature-set of web content.
- JavaScript is a scripting language that runs in your web browser (and also outside of the browser), and can implement complex features on web pages — for example, it can update the content displayed on a page in response to user actions, or load new content from a server and display it on the page, or even create a complete client-side application that runs in a browser. JavaScript is also used in many non-browser environments, such as Node.js, Electron, and Cordova. JavaScript is a very powerful language, and it's often used to implement "web effects" — for example, it can be used to implement some simple animations, or to load and display data from a third-party API. JavaScript is also used in many non-web projects, such as Node.js, Electron, and Cordova.

```
Player 1: Chris
```

```
let myObject = {type: "Player", name: "Chris"};
```

Then we can use them in JS like this to get something out:

```
console.log(myObject.name);
```

```
Player 1: Chris
```

And finally, we can add some JavaScript to implement operations between them:

```
function updateName() {
  let myName = prompt('Please enter your name.');
  if (!myName || myName === null) {
    let myStore = localStorage.getItem('name');
    if (myStore && myStore !== null) {
      myName = myStore;
    } else {
      myName = prompt('Sorry, we need your name.');
    }
  }
  localStorage.setItem('name', myName);
  let myHeader = document.getElementById('header');
  myHeader.textContent = 'Welcome, ' + myName;
}
let myButton = document.getElementById('button');
myButton.addEventListener('click', updateName);
```

```
Welcome, Chris
```

You can also "Tweak" the code and see the example in the MDN Playground. Try clicking on the text you see in the output.

```
console.log('Hello, world!');
```

```
Hello, world!
```

JavaScript can do a lot more than that — it can respond to what is in store data:

```
let myStore = localStorage.getItem('name');
if (myStore && myStore !== null) {
  myName = myStore;
}
```

```
Welcome, Chris
```